Death Avoidance as the Dominant Routing Signal: A Counterfactual Analysis on ScienceWorld

Anonymous ACL submission

Abstract

Modular LLM agents that route between neural reasoning and deterministic computation promise improved reliability, yet understanding *what* drives the value of routing decisions remains unclear. We propose episode-level counterfactual analysis: running complete episodes under pure strategies and comparing final outcomes to decompose routing signal into death-avoidance and non-death components. Among task types exhibiting meaningful routing signal in prescreening (11 of 30 available ScienceWorld task types, 245 episodes, Qwen2.5-7B-Instruct), 78.9% of non-tie routing labels (75 of 95) are driven by death asymmetry—a penalty-independent metric. Under the default penalty P=100, the Death Avoidance Factor (DAF) is 90.4% (95% CI: $[81.6\%, 98.8\%]$), though DAF varies from 50% (P=10) to 98% (P=500) across penalty settings (Appendix F). Per-task death rate correlates strongly with routing signal (r=0.97, though partly structural as death penalties contribute directly to score differences; §3.1). An exploratory step-level router trained on these labels achieves $54.1 \pm 5.4\%$ step-level balanced accuracy (with same-episode leakage) but only $47.1 \pm 2.2\%$ on held-out episodes (below the always-deterministic baseline of 50%); data scaling ($32\times$) confirms this capacity-dependent limitation; scaling to 4B yields 59.0% episode-level accuracy ($p < 0.005$ vs. random) but the cost (4B parameters for $+9$ pp) is disproportionate to the routing benefit. Cross-environment analysis on MATH (N=300) shows the death-driven routing signal is not statistically detectable without a death penalty: the label distribution inverts from 86:9 (deterministic-preferred) to 30:17 (internal directionally favored), with a directional trend favoring internal reasoning ($+4.3$ percentage points, $p = 0.079$, not reaching α=0.05). A Llama-3.1-8B-Instruct replication (7 task types, 70 episodes) shows direction-consistent death patterns for all task types, with a larger death-rate differential ($+34.3\%$ vs. $+26.6\%$ for Qwen) and zero internal-preferred episodes. These findings indicate that in environments with catastrophic failure modes, routing value is dominated by safety—the routing problem reduces to safety-first modality selection rather than efficiency optimization.

1 Introduction

Large language models deployed as interactive agents must repeatedly decide *how* to compute: reason internally, or delegate to an external tool. A Qwen2.5-7B-Instruct agent navigating ScienceWorld (Wang et al., 2022)—a text-based science simulation with 30 task types—faces this choice at every step. Tracking temperature thresholds, container states, and multi-step procedures via internal reasoning is error-prone: the model may hallucinate that water has boiled when the stove was never activated, or that an object has been moved when the action was invalid. Deterministic external computation (environment simulators, code interpreters) eliminates such hallucinations via exact state transitions. The practical question for agent design is therefore: *when should an agent route computation externally versus reasoning internally?*

Existing routing approaches frame this as a capability or efficiency decision. Toolformer (Schick et al., 2023) teaches models to insert API calls when doing so reduces perplexity; ART (Paranjape et al., 2023) decomposes tasks into tool-augmented programs; Chameleon (Lu et al., 2023) selects among heterogeneous modules (search, code, vision) per subtask. More recent adaptive-computation methods route to different reasoning depths (Yang et al., 2026b,a) or model tiers (Zhang et al., 2026b). The shared assumption is that routing signal stems from diverse sources—accuracy, latency, capability matching—and that a learned router can exploit this signal to improve over any fixed strategy. But this assumption has never been

systematically verified: *what actually drives the value of routing decisions in interactive environments?*

We find a surprisingly concentrated answer: **death avoidance accounts for 90.4% of the routing signal** (DAF $= 90.4\%$, 95% CI: $[81.6\%, 98.8\%]$; at default penalty P=100) in ScienceWorld. The internal strategy's hallucination-induced failures trigger catastrophic agent death (a -100 penalty), while the deterministic strategy survives the same episodes. Once death episodes are removed, the performance gap between strategies collapses to $\Delta_{\text{nd}} = +2.84$.

Reaching this conclusion requires a methodological contribution. The standard approach to generating routing labels—per-step oracle comparison—degenerates under sparse rewards. In ScienceWorld, non-zero reward arrives only at episode termination; 97.4% of steps (across 11 prescreened task types) produce identical scores regardless of modality, leaving no step-level signal to learn from (§2.2). Keyword-based heuristics also fail structurally, as ScienceWorld actions lack science-specific terminology (§3.2). We instead propose *episode-level counterfactual analysis*: run complete episodes under pure-internal (ReAct (Yao et al., 2023b)) and pure-deterministic strategies, then compare episode outcomes to generate routing labels (§2.2).

Among task types exhibiting meaningful routing signal in prescreening (11 of 30 available ScienceWorld task types, 245 episode pairs), the full deterministic advantage is $\Delta_{\text{full}} = +29.60$ points. Excluding death episodes, the gap collapses to $\Delta_{\text{nd}} = +2.84$; 78.9% of non-tie labels (75 of 95) arise from death asymmetry—a penalty-independent measure. The penalty-sensitive Death Avoidance Factor (DAF; §2.4) is 90.4% (95% CI: $[81.6\%, 98.8\%]$) at the default penalty P=100, ranging from 50% (P=10) to 98% (P=500). The per-task correlation between death-rate differential and performance gap is $r = 0.97$ ($p < 10^{-6}$, $n = 11$), confirming that death avoidance is the dominant mechanism across task types (§3.1).

Two cross-context checks probe the specificity of this finding. First, a MATH Level 4–5 evaluation (300 problems), using a code interpreter as the deterministic modality, shows a directional trend consistent with the absence of death-driven routing signal: internal reasoning outperforms code interpreter by $+4.3$ pp ($p = 0.079$, not reaching α=0.05)—the opposite direction from ScienceWorld. In the absence of death risk, the deterministic advantage is not statistically detectable ($p = 0.079$, not reaching α=0.05). Second, a cross-model replication with Llama-3.1-8B-Instruct (7 task types, 70 episode pairs) shows direction-consistent death patterns for all task types, with a larger death-rate differential ($+34.3\%$ vs. $+26.6\%$ for Qwen; Appendix G), confirming the phenomenon generalizes across model families. An exploratory step-level router (Qwen3-1.7B with LoRA) trained on these counterfactual labels achieves $54.1 \pm 5.4\%$ step-level balanced accuracy but only $47.1 \pm 2.2\%$ on held-out episodes—below the always-deterministic baseline—a capacity-dependent limitation confirmed by data scaling ($32\times$) ablations; the 4B scale yields 59.0% but the cost is disproportionate to the routing benefit (§3.3).

Our contributions are:

1. **Routing signal decomposition.** We identify step-level oracle degeneracy in sparse-reward environments and introduce episode-level counterfactual analysis with the DAF metric, a methodology for quantifying what fraction of routing signal derives from catastrophic-failure avoidance (§2.2–2.4).

2. **Death avoidance dominance.** On ScienceWorld, 78.9% of non-tie routing labels stem from death asymmetry. The penalty-dependent DAF metric reaches 90.4% at P=100 but varies substantially across penalty settings. An exploratory router trained on these labels fails to outperform the trivial always-deterministic baseline on held-out episodes ($47.1 \pm 2.2\%$ vs. 50.0%); data scaling ablations confirm this capacity-dependent limitation; scaling to 4B yields 59.0% episode-level accuracy but the cost (4B parameters for $+9$ pp above random) is disproportionate when DAF $= 90.4\%$ and always-deterministic is trivially near-optimal (§3.1–3.3).

3. **Cross-context validation.** On MATH (N=300), where no death penalty exists, internal reasoning directionally outperforms code interpretation ($+4.3$ pp, $p = 0.079$, not reaching α=0.05), consistent with the absence of death-driven routing signal. On ALFWorld (N=135), deterministic success (27.4%) far exceeds internal (3.0%), but the mechanism

is information asymmetry (access to admissible actions) rather than death avoidance (§5). Both contrasts confirm that the ScienceWorld finding is driven by failure structure, not an inherent advantage of deterministic computation.

2 Method

2.1 Problem Setup

Consider an interactive environment $\mathcal{E}$ with state space $\mathcal{S}$, observation space $\mathcal{O}$, and action space $\mathcal{A}$. An agent follows a policy $\pi : \mathcal{O}^* \to \mathcal{A}$ that maps observation histories to actions. At each step t, the agent receives observation o_t, selects action $a_t = \pi(o_1, \ldots, o_t)$, and the environment transitions to state s_{t+1}.

We define a modality set $\mathcal{M} = \{\texttt{int}, \texttt{det}\}$ governing how the agent processes state transitions:

- **Internal** ($m = \texttt{int}$): The agent uses its own generation to predict intermediate observations. The LLM interleaves chain-of-thought reasoning with environment actions following the ReAct paradigm (Yao et al., 2023b), but prediction errors compound across steps.
- **Deterministic** ($m = \texttt{det}$): The agent receives ground-truth observations from the environment simulator at each step—equivalent to access to a perfect dynamics model. Action selection remains with the LLM; only the state-transition prediction is replaced.

We restrict $\mathcal{M}$ to two modalities. An ablation comparing chain-of-thought reasoning (deep internal) against direct-answer generation (fast internal) on 6 overlapping task types (n=120 episodes per condition) found no significant difference on any task (Mann–Whitney $p > 0.26$, $|d| \leq 0.31$) and near-identical death rates (41.8% vs. 42.5%), confirming that reasoning depth does not meaningfully alter internal-mode outcomes in this setting.

A *routing policy* $\rho : \mathcal{O}^* \to \mathcal{M}$ selects which modality to use at each step. The goal of modality routing is to learn ρ that maximizes expected episode return. Our analysis investigates what signal is available to learn such a router.

2.2 Counterfactual Episode Collection

Per-step oracle labeling assigns routing labels by comparing single-step outcomes under each computation modality. Under sparse terminal rewards, this approach degenerates: on ScienceWorld (Wang et al., 2022), 97.4% of steps (across 11 prescreened task types) produce identical intermediate scores regardless of modality, yielding no routing signal. We instead perform episode-level counterfactual analysis.

Pure-strategy trajectories. For a task instance t and modality $m \in \mathcal{M}$, a *pure-strategy trajectory* is the complete episode obtained by applying modality m at every step:

$$\tau_m^{(t)} = (o_1, a_1, o_2, a_2, \ldots, o_T, a_T) \tag{1}$$

where $T \leq T_{\max}$ is the episode length (capped at $T_{\max} = 30$ steps). Both trajectories $\tau_{\texttt{int}}^{(t)}$ and $\tau_{\texttt{det}}^{(t)}$ are executed under identical initial conditions (same random seed, same task variation), ensuring that performance differences arise solely from the modality.

Each trajectory τ yields a terminal score $s(\tau) \in \mathbb{Z}$ and a death indicator $d(\tau) \in \{0, 1\}$. When the agent dies ($d = 1$), the environment imposes a penalty: $s(\tau) = -P$ where $P > 0$ ($P = 100$ in ScienceWorld). The episode pair produces a tuple:

$$\begin{aligned}\big(s^{\texttt{int}}, s^{\texttt{det}}, d^{\texttt{int}}, d^{\texttt{det}}\big) \triangleq \\ \big(s(\tau_{\texttt{int}}),\, s(\tau_{\texttt{det}}),\, d(\tau_{\texttt{int}}),\, d(\tau_{\texttt{det}})\big)\end{aligned} \tag{2}$$

We use Qwen2.5-7B-Instruct (vLLM) on ScienceWorld (Wang et al., 2022), a text-based environment spanning 30 task types with a -100 death penalty. After prescreening, we collect 245 episode pairs across 11 task types (20–40 per type). The framework is environment-agnostic: it requires only paired episode outcomes under two modalities, with no assumptions about score structure or failure mode. We apply it unchanged to ALFWorld and MATH in §5.

2.3 Episode-Level Labeling

We define a labeling function L that maps a counterfactual pair to a routing preference:

$$\begin{aligned}L\big(\tau_{\texttt{int}}^{(t)}, \tau_{\texttt{det}}^{(t)}\big) \;\to \\ \ell \in \{\texttt{det-pref},\ \texttt{int-pref},\ \texttt{no-pref}\}\end{aligned} \tag{3}$$

The labeling function is *death-aware*: it prioritizes survival over score, reflecting the asymmetric cost of catastrophic failure in interactive environments. ScienceWorld scores are integer-valued, so we use exact (in)equality with no threshold. The

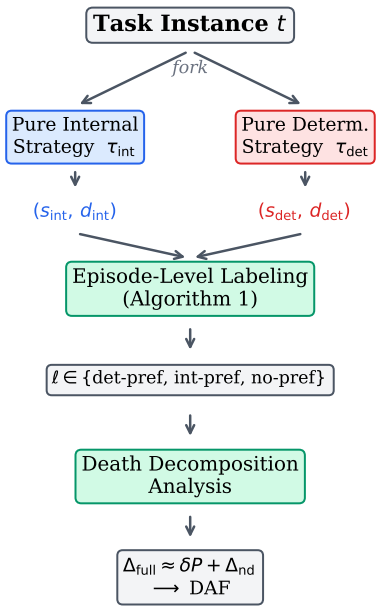


Figure 1: Counterfactual episode collection and analysis pipeline. Each task instance is run under both pure strategies independently, producing episode-level labels via Algorithm 1.

Algorithm 1 Episode-Level Counterfactual Labeling

Require: Task instance t, modalities $\mathcal{M} = \{\texttt{int}, \texttt{det}\}$
Ensure: Label $\ell \in \{\texttt{det-pref}, \texttt{int-pref}, \texttt{no-pref}\}$

1: **for** each $m \in \mathcal{M}$ **do**
2: Run episode τ_m with pure-strategy m
3: Record score s_m and death indicator d_m
4: **end for**
5: **if** $d_{\texttt{int}} = 1$ **and** $d_{\texttt{det}} = 0$ **then**
6: $\ell \leftarrow \texttt{det-pref}$ {death asymmetry}
7: **else if** $d_{\texttt{int}} = 0$ **and** $d_{\texttt{det}} = 1$ **then**
8: $\ell \leftarrow \texttt{int-pref}$ {death asymmetry}
9: **else if** $d_{\texttt{int}} = d_{\texttt{det}} = 1$ **then**
10: $\ell \leftarrow \texttt{no-pref}$ {both dead}
11: **else if** $s_{\texttt{det}} > s_{\texttt{int}}$ **then**
12: $\ell \leftarrow \texttt{det-pref}$ {score comparison}
13: **else if** $s_{\texttt{int}} > s_{\texttt{det}}$ **then**
14: $\ell \leftarrow \texttt{int-pref}$ {score comparison}
15: **else**
16: $\ell \leftarrow \texttt{no-pref}$ {tied alive}
17: **end if**
18: **return** ℓ

full procedure is given in Algorithm 1. Figure 1 illustrates the overall pipeline.

The four mutually exclusive cases correspond to:

1. **Death asymmetry** (lines 5–8): $d^{\texttt{int}} \neq d^{\texttt{det}}$ → the surviving strategy is preferred. This accounts for 78.9% of all non-tie labels (75 of 95).

2. **Both dead** (lines 9–10): $d^{\texttt{int}} = d^{\texttt{det}} = 1$ → `no_preference`.

3. **Score comparison** (lines 11–14): both survive and $s^{\texttt{det}} \neq s^{\texttt{int}}$ → the higher-scoring strategy is preferred.

4. **Tied alive** (lines 15–16): both survive and $s^{\texttt{det}} = s^{\texttt{int}}$ → `no_preference`.

Label distributions are reported in §3.1.

2.4 Death Avoidance Factor

To quantify the contribution of death avoidance to routing signal, we decompose the mean score gap between strategies. Let $\delta = \mathbb{E}[d^{\texttt{int}}] - \mathbb{E}[d^{\texttt{det}}]$ denote the death-rate differential and Δ_{nd} the mean score gap restricted to episodes where neither strategy dies. The full gap decomposes as:

$$\Delta_{\text{full}} \approx \underbrace{\delta \cdot P}_{\text{death contribution}} + \Delta_{\text{nd}} \tag{4}$$

The approximation holds exactly when the surviving strategy scores zero in all death-asymmetry episodes; in our data, the residual is small ($29.60 - 29.37 = 0.23$, or 0.8% of Δ_{full}).

We define the Death Avoidance Factor (DAF) as the fraction of the full score gap attributable to death avoidance:

$$\text{DAF} = 1 - \frac{\Delta_{\text{nd}}}{\Delta_{\text{full}}} \approx \frac{\delta P}{\delta P + \Delta_{\text{nd}}} \tag{5}$$

[1]

DAF is well-defined when $\Delta_{\text{full}} > 0$; in our data, the minimum per-task Δ_{full} is $+9.58$ (`find-non-living-thing`), ensuring numerical stability. Since $\text{DAF} = \delta P / (\delta P + \Delta_{\text{nd}})$, it increases monotonically with P—a mathematical consequence, not an empirical finding. We

[1] When $\Delta_{\text{nd}} < 0$ (deterministic strategy scores lower than internal on non-death episodes), DAF exceeds 1, indicating death avoidance overcompensates for a non-death disadvantage.

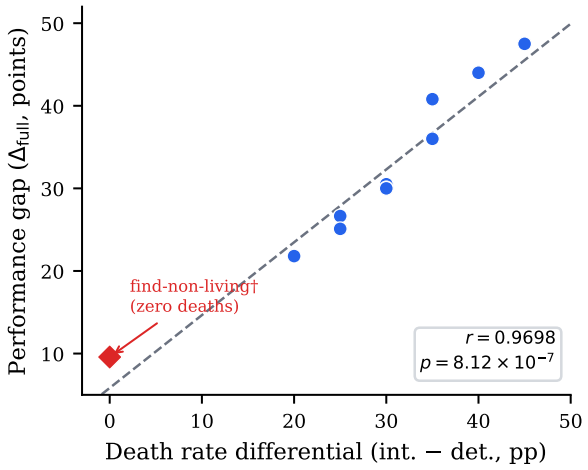


Figure 2: Per-task scatter: death rate differential (internal − deterministic) vs. performance gap (Δ_{full}). $r = 0.97$, $p < 10^{-6}$, $n = 11$ task types.

therefore report both DAF at the environment default (P=100) and the penalty-independent death-asymmetry label fraction.

To assess sensitivity to penalty magnitude, we recalculate DAF under hypothetical death penalties (Appendix F): P=10 → DAF = 50.4%, P=50 → 82.6%, P=100 → 90.4%, P=200 → 95.0%. Death avoidance remains the dominant component across all penalty magnitudes, though its relative contribution scales with penalty severity. As a penalty-independent companion metric, 78.9% of non-tie routing labels (75 of 95) are driven by death asymmetry rather than score comparison (§2.3), confirming that death avoidance dominates routing signal regardless of penalty magnitude.

3 Experiments

3.1 Death Avoidance Decomposition

Across 245 episodes, $\Delta_{\text{full}} = +29.60$; excluding death episodes, $\Delta_{\text{nd}} = +2.84$, yielding DAF = 90.4% (95% CI: [81.6%, 98.8%]; at default P=100). The penalty-independent death-asymmetry label fraction is 78.9% (75 of 95 non-tie labels). The internal strategy dies in 42.9% of episodes versus 16.3% for deterministic (2.6×). Per-task, death rate differential correlates strongly with Δ_{full} ($r = 0.97$, $p < 10^{-6}$; Figure 2; robustness diagnostics in Appendix C). A high correlation is expected whenever the scaled death-rate differential ($P \times \Delta_d$) dominates between-task variance—formally, when $\text{Var}(P \cdot \Delta_d) \gg \text{Var}(\Delta_{\text{nd}})$. At P=100, this condition holds for all but one task type (`find-non-living-thing`, which has zero deaths). The sensitivity analysis (Appendix F) confirms that r drops below significance at $P \leq 25$, where this dominance condition no longer holds. Of 245 episodes, 86 are `det_preferred`, 9 `internal_preferred`, and 150 `no_preference`.

One exception: `find-non-living-thing` (n=40, zero deaths) has DAF = 0% with $\Delta = +9.58$ entirely from non-death score differences; all 10 remaining types exceed DAF > 85%. Bootstrap 95% CIs confirm $\Delta > 0$ for 10 of 11 types; `chemistry-mix` (N=20, $\Delta = 21.8$, 95% CI $[-5.0, 48.6]$) is the sole exception (Appendix B).

Strategy	Score	95% CI	Death %
Always-Internal	−39.07	[−46.40, −32.27]	42.9
Always-Deterministic	−9.47	[−14.71, −4.29]	16.3
Random	−22.38	[−28.67, −15.75]	27.8
Oracle	−6.58	[−11.54, −1.70]	14.3
Trained Router†	−10.70	[−16.33, −5.27]	17.6

Table 1: Episode-weighted mean scores on ScienceWorld (11 tasks, 245 episodes). † Single seed; see §3.3.

3.2 Baselines

Always-deterministic (−9.47) outperforms always-internal (−39.07) by 29.60 points. The oracle improves only to −6.58, with 96.3% of decisions matching always-deterministic—*oracle degeneracy* indicating the optimal routing policy closely resembles unconditional deterministic computation.

3.3 Router Training as Diagnostic

We probe whether counterfactual labels can train a step-level router, as a diagnostic of label informativeness. A classification head on LLM hidden states degenerates to always-deterministic across all configurations. An implicit router (Qwen3-1.7B fine-tuned; Appendix D) yields 54.1 ± 5.4% step-level balanced accuracy but 47.1 ± 2.2% on held-out episodes (8 seeds)—below always-deterministic (50%). This reveals a capacity-dependent negative: at practical router scales (0.6B–1.7B), episode-level balanced accuracy remains at or below chance (50.67%/47.1%). Scaling to 4B yields 58.96 ± 1.94% ($p < 0.005$ vs. random; Appendix I)—confirming that episode-level signal exists but is capacity-starved. Data scaling (25%–800%) shows no improvement at any capacity (Appendix E). Notably, scaling model capacity improves episode-level routing (+9 pp at 4B) without improving step-level accuracy (flat ∼53%

across scales), confirming that the extractable signal is task-type-level structure—consistent with §3.1's finding that death risk (a task-level property) dominates routing utility. The cost of this extraction (4B parameters for $+9$ pp) is disproportionate to the routing benefit, validating our analytical methodology as the efficient alternative to brute-force learned routing. Deployed end-to-end (Table 1), the router (-10.70) does not outperform always-deterministic (-9.47; overlapping CIs). We verify label quality in Appendix A.

A Llama-3.1-8B-Instruct replication (7 task types, N=70) confirms direction-consistent death patterns for all tasks, with a larger differential ($+34.3\%$ vs. $+26.6\%$; Appendix G). Failure mode and ablation details in Appendices H–I.

4 Related Work

Tool Use and External Computation in LLMs. A growing body of work augments language models with external tools (Qu et al., 2024). Toolformer (Schick et al., 2023) teaches LLMs to insert API calls via self-supervised loss filtering; ToolkenGPT (Hao et al., 2023) represents tools as special tokens enabling frozen models to invoke massive tool libraries; ART (Paranjape et al., 2023) decomposes tasks into tool-augmented programs; Chameleon (Lu et al., 2023) composes heterogeneous modules per subtask; ReTool (Feng et al., 2025) applies RL to discover when code execution is preferable to chain-of-thought. These focus on *learning to use tools*; we ask a prior question: *what signal drives the routing decision?* Recent concurrent work sharpens this question: Zeng et al. (2026) identify a cognitive-illusion effect in LLM tool use, Sun et al. (2026) show tool-call decisions are encoded in hidden states before reasoning, Zhang et al. (2026a) quantify a "tool-use tax" where tools degrade performance, and Wu et al. (2026) propose a necessity–utility–affordability framework. Our counterfactual decomposition provides environment-grounded evidence: in ScienceWorld, 90.4% of routing signal at P=100 traces to catastrophic-failure prevention.

Adaptive Computation and Routing. CogRouter (Yang et al., 2026b) adapts reasoning depth per step, ARES (Yang et al., 2026a) selects among reasoning effort levels, Mixture-of-Recursions (Bae et al., 2025) learns token-level recursion depths, and MTRouter (Zhang et al., 2026b) addresses multi-turn routing across heterogeneous models. RouteLLM (Ong et al., 2024) learns routers from preference data to select between strong and weak models. These optimize *how* to route; we reveal *why* routing helps—in interactive environments, the dominant signal is catastrophic-failure avoidance.

Counterfactual Evaluation in RL. Prior work generates counterfactual explanations for RL agents by perturbing individual actions within a fixed policy (Olson et al., 2021; Gajcin, 2023; Amitai et al., 2024). Our approach differs in both purpose and granularity: we execute two complete episodes under different routing strategies, producing prospective paired comparisons that isolate routing signal *before* any router is trained, sidestepping compounding perturbation artifacts.

Safe RL and Catastrophic Failure. García and Fernández (2015) survey safe RL methods that constrain policies to avoid catastrophic outcomes; Xu et al. (2021) develop principled constrained optimization with formal convergence guarantees. Our counterfactual decomposition reveals that death avoidance is the dominant component of routing signal, suggesting that modality routing functions primarily as a safety mechanism rather than a generic capability-matching problem.

LLM Agents in Interactive Environments. ScienceWorld (Wang et al., 2022) serves as our primary testbed; ALFWorld (Shridhar et al., 2021) provides a complementary household environment; Voyager (Wang et al., 2023) demonstrates open-ended exploration in Minecraft using LLM-driven curriculum. ReAct (Yao et al., 2023b) interleaves reasoning with actions and serves as our internal baseline; Reflexion (Shinn et al., 2023) extends this with verbal self-reflection. Recent agent architectures augment planning with world models (Liu et al., 2026; Yu et al., 2025), which inherit hallucination risks. Our deterministic modality provides exact state transitions, and our analysis shows this exactness matters almost exclusively for death avoidance.

5 Discussion

Cross-environment evidence. Does routing signal persist without catastrophic failure? On MATH Level 4–5 (300 problems), using a code interpreter as the deterministic modality, internal reasoning directionally outperforms: 81.0% vs. 76.7% accuracy ($+4.3$ pp, 95% CI $[0.0, +8.7]$; McNemar $p = 0.079$, not reaching α=0.05). The label

distribution inverts: 30 internal-preferred vs. 17 deterministic-preferred (cf. 9:86 in ScienceWorld), with 84.3% no-preference. In the absence of death risk, the death-driven routing signal is not statistically detectable—consistent with death avoidance as the primary driver. A complementary evaluation on ALFWorld (N=135) confirms that the counterfactual framework identifies different routing mechanisms across environments: the deterministic advantage there stems from information asymmetry (access to admissible actions) rather than death avoidance (Appendix J).

Penalty sensitivity. The $r = 0.97$ correlation is specific to catastrophic death (score $= -100$); defining "failure" as any non-positive score yields $r = -0.13$ ($p = 0.70$). The correlation emerges only at $P \geq 50$ ($r = 0.86$) and reflects that non-death episodes show near-zero performance differential ($\Delta_{\text{nd}} = 2.84$; Appendix F). Death is effectively the *only* component generating differential routing signal in ScienceWorld. These findings may be capability-dependent: the 42.9% internal death rate reflects a 7B model's hallucination vulnerability; larger models with lower death rates would reduce DAF, potentially revealing non-death routing signal currently masked by catastrophic failures.

Implications for router design. The dominance of death avoidance (DAF $= 90.4\%$ at P=100) reframes modality routing as primarily a *safety* problem rather than an efficiency-quality tradeoff. The routing problem reduces to safety-first modality selection—closer to a conservative default policy than to a learned safety critic—aligning more closely with safe RL (García and Fernández, 2015) than with cost-quality optimization (Yang et al., 2026b,a). The extreme label imbalance (86:9 deterministic-preferred, 150 no-preference) is not an artifact but a central result: among episodes where modality matters, deterministic is preferred at 9.6:1, explaining why always-deterministic is a strong baseline and why the trained router (§3.3) adds no value. A lightweight death predictor may capture most routing benefit without full counterfactual labeling. The capacity scaling results reinforce this: extracting episode-level routing signal requires 4B parameters for only $+9$ pp above random (59.0%), a disproportionate cost that validates our counterfactual analytical methodology as the efficient alternative to brute-force learned routing. In principle, routing could be fully reduced to per-task-type death-risk prediction, bypassing step-level observation entirely—a direction we leave for future work.

Modality heterogeneity. The deliberate heterogeneity in deterministic modality definition—ground-truth state observation in ScienceWorld, code execution in MATH, admissible-action filtering in ALFWorld—tests whether the framework (episode-level counterfactual analysis) generalizes, not whether any single modality definition does. The consistent finding across structurally different modalities strengthens rather than weakens the conclusion that catastrophic-failure avoidance is the dominant routing signal wherever it exists.

6 Conclusion

We introduced episode-level counterfactual analysis for decomposing routing signal in modular LLM agents. Applied to 245 ScienceWorld episodes (11 of 30 task types, Qwen2.5-7B-Instruct), 78.9% of non-tie routing labels (75 of 95) arise from death asymmetry. The DAF reaches 90.4% (95% CI: $[81.6\%, 98.8\%]$) at P=100, with strong cross-task correlation ($r = 0.97$, $p < 10^{-6}$).

Cross-environment evidence supports this finding: on MATH (N=300), the death-driven routing signal is not statistically detectable without death penalty ($+4.3$ pp, $p = 0.079$, not reaching α=0.05). A Llama-3.1-8B-Instruct replication (7 task types, 70 episodes) confirms direction-consistent death patterns for all tasks with a larger differential ($+34.3\%$ vs. $+26.6\%$). An exploratory router trained on counterfactual labels achieves only $47.1 \pm 2.2\%$ on held-out episodes—below always-deterministic—a capacity-dependent limitation confirmed by data scaling; the 4B scale yields 59.0% episode-level accuracy but the cost is disproportionate to the routing benefit.

In environments with catastrophic failure modes, routing value is dominated by safety: the routing problem reduces to safety-first modality selection rather than efficiency optimization. Open directions include step-level counterfactual labeling, stronger backbones with lower intrinsic death rates, and environments with graded failure costs.

Limitations

Our primary analysis uses a single environment (ScienceWorld, 11 of 30 task types) with one backbone (Qwen2.5-7B-Instruct); generalization requires further validation. The -100 death penalty

amplifies DAF; environments with softer penalties would exhibit lower DAF values (Appendix F). The 19 excluded task types (63%) lack routing signal because both strategies typically time out within 30 steps; increasing the step budget could surface additional signal. Our internal-reasoning baseline uses single-pass ReAct (Yao et al., 2023b); more sophisticated strategies (e.g., Reflexion (Shinn et al., 2023) or Tree-of-Thought backtracking (Yao et al., 2023a)) may reduce internal death rates and lower DAF. Episode-level labels may assign suboptimal routing to individual steps, particularly those preceding death events, and training on pure-strategy trajectories introduces distribution shift at deployment (Ross et al., 2011). Cross-environment analyses on MATH and ALFWorld (§5, Appendix J) provide supporting evidence but do not conclusively establish generalization. A Llama-3.1-8B-Instruct replication (7 task types, N=70; Appendix G) confirms direction-consistent death patterns for all tasks, though both models are in the 7–8B parameter range; validation at larger scales remains future work.

Ethics Statement

This work uses ScienceWorld (Wang et al., 2022), a publicly available text-based simulation environment, and does not involve human subjects or private data. All language models are accessed via standard APIs or publicly released weights. We report computational costs: approximately 40 GPU-hours on two consumer GPUs for data collection, router training, and evaluation. Code and data will be released upon acceptance.

References

Yotam Amitai, Guy Septon, and Ofra Amir. 2024. Explaining reinforcement learning agents through counterfactual action outcomes. In *Proceedings of the AAAI Conference on Artificial Intelligence*.

Sangmin Bae and 1 others. 2025. Mixture-of-recursions: Learning dynamic recursive depths for adaptive token-level computation. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Jiazhan Feng and 1 others. 2025. Retool: Reinforcement learning for strategic tool use in llms. *arXiv preprint arXiv:2504.11536*.

Jasmina Gajcin. 2023. Counterfactual explanations for reinforcement learning agents. In *Proceedings of the International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*.

Javier García and Fernando Fernández. 2015. A comprehensive survey on safe reinforcement learning. *Journal of Machine Learning Research*, 16(1):1437–1480.

Shibo Hao, Tianyang Liu, Zhen Wang, and Zhiting Hu. 2023. Toolkengpt: Augmenting frozen language models with massive tools via tool embeddings. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Youwei Liu and 1 others. 2026. Imagine-then-plan: Agent learning from adaptive lookahead with world models. *arXiv preprint arXiv:2601.08955*.

Pan Lu, Baolin Peng, Hao Cheng, Michel Galley, Kai-Wei Chang, Ying Nian Wu, Song-Chun Zhu, and Jianfeng Gao. 2023. Chameleon: Plug-and-play compositional reasoning with large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Matthew L. Olson, Roli Khanna, Lawrence Neal, Fuxin Li, and Weng-Keen Wong. 2021. Counterfactual state explanations for reinforcement learning agents via generative deep learning. *Artificial Intelligence*, 295:103455.

Isaac Ong, Amjad Almahairi, Vincent Wu, Wei-Lin Chiang, Tianhao Wu, Joseph E. Gonzalez, M Waleed Kadous, and Ion Stoica. 2024. RouteLLM: Learning to route LLMs with preference data. *arXiv preprint arXiv:2406.18665*.

Bhargavi Paranjape, Scott Lundberg, Sameer Singh, Hannaneh Hajishirzi, Luke Zettlemoyer, and Marco Tulio Ribeiro. 2023. Art: Automatic multi-step reasoning and tool-use for large language models. *arXiv preprint arXiv:2303.09014*.

Changle Qu, Sunhao Dai, Xiaochi Wei, Hengyi Cai, Shuaiqiang Wang, Dawei Yin, Jun Xu, and Ji-Rong Wen. 2024. Tool learning with large language models: A survey. *arXiv preprint arXiv:2405.17935*.

Stéphane Ross, Geoffrey J Gordon, and Drew Bagnell. 2011. A reduction of imitation learning and structured prediction to no-regret online learning. In *Proceedings of the International Conference on Artificial Intelligence and Statistics (AISTATS)*.

Timo Schick, Jane Dwivedi-Yu, Roberto Dessì, Roberta Raileanu, Maria Lomeli, Eric Hambro, Luke Zettlemoyer, Nicola Cancedda, and Thomas Scialom. 2023. Toolformer: Language models can teach themselves to use tools. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. 2023. Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Mohit Shridhar, Xingdi Yuan, Marc-Alexandre Côté, Yonatan Bisk, Adam Trischler, and Matthew Hausknecht. 2021. Alfworld: Aligning text and embodied environments for interactive learning. In *International Conference on Learning Representations (ICLR)*.

Chung-En Sun, Linbo Liu, Ge Yan, Zimo Wang, and Tsui-Wei Weng. 2026. LLM agents already know when to call tools – even without reasoning. *arXiv preprint arXiv:2605.09252*.

Guanzhi Wang, Yuqi Xie, Yunfan Jiang, Ajay Mandlekar, Chaowei Xiao, Yuke Zhu, Linxi Fan, and Anima Anandkumar. 2023. Voyager: An open-ended embodied agent with large language models. *arXiv preprint arXiv:2305.16291*.

Ruoyao Wang, Peter Jansen, Marc-Alexandre Côté, and Prithviraj Ammanabrolu. 2022. Scienceworld: Is your agent smarter than a 5th grader? In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

Qinyuan Wu, Soumi Das, Mahsa Amani, Arijit Nag, Seungeon Lee, Krishna P. Gummadi, Abhilasha Ravichander, and Muhammad Bilal Zafar. 2026. To call or not to call: A framework to assess and optimize LLM tool calling. *arXiv preprint arXiv:2605.00737*.

Tengyu Xu, Yingbin Liang, and Guanghui Lan. 2021. CRPO: A new approach for safe reinforcement learning with convergence guarantee. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*.

Jingbo Yang and 1 others. 2026a. Ares: Adaptive reasoning effort selection for efficient llm agents. *arXiv preprint arXiv:2603.07915*.

Ruihan Yang and 1 others. 2026b. Think fast and slow: Step-level cognitive depth adaptation for llm agents. *arXiv preprint arXiv:2602.12662*.

Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Thomas L Griffiths, Yuan Cao, and Karthik Narasimhan. 2023a. Tree of thoughts: Deliberate problem solving with large language models. In *NeurIPS*.

Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2023b. React: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations (ICLR)*.

Xiao Yu and 1 others. 2025. Dyna-think: Synergizing reasoning, acting, and world model simulation in ai agents. *arXiv preprint arXiv:2506.00320*.

Yirong Zeng, Shen You, Yufei Liu, Qunyao Du, Xiao Ding, Yutai Hou, Yuxian Wang, Wu Ning, Haonan Song, Dandan Tu, Bibo Cai, and Ting Liu. 2026. The tool-overuse illusion: Why does LLM prefer external tools over internal knowledge? *arXiv preprint arXiv:2604.19749*.

Kaituo Zhang, Zhen Xiong, Mingyu Zhong, Zhimeng Jiang, Zhouyuan Yuan, Zhecheng Li, and Ying Lin. 2026a. Are tools all we need? unveiling the tool-use tax in LLM agents. *arXiv preprint arXiv:2605.00136*.

Yiqun Zhang and 1 others. 2026b. Mtrouter: Cost-aware multi-turn llm routing with history-model joint embeddings. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics (ACL)*.

A Labeling Quality Audit

A potential concern is oracle circularity: whether episode-level labels reflect tie-breaking artifacts rather than genuine modality differences. We audit all 245 labels. Of the 150 `no_preference` labels (61.2%), 115 (46.9%) are both-alive ties and 35 (14.3%) are both-dead—all excluded from router training. The remaining 95 training labels are entirely genuine: 75 from death asymmetry (70 deterministic-preferred, 5 internal-preferred) and 20 from score comparison with both strategies surviving (16 deterministic-preferred, 4 internal-preferred). Zero labels depend on tie-breaking. Retraining the router using only death-asymmetry labels yields step-level balanced accuracy of 57.3%, within 1σ of the full-label retrain ($57.6 \pm 0.5\%$, step-level Trainer evaluation), confirming that score-comparison labels contribute negligible additional signal. (Step-level metrics are used here for the relative comparison; as shown in §3.3, absolute generalization to new episodes is near chance.) The 97.4% step-level tie rate (§2.2) further motivates episode-level over step-level labeling.

B Per-Task Decomposition

We select 11 of ScienceWorld's 30 task types via prescreening: task types where both strategies complete at least one episode within the 30-step budget are retained; types where both strategies time out on all seed variations are excluded.[2]

Task selection and anti-cherry-picking. The 11 included task types are those exhibiting meaningful routing signal in prescreening, not those with the highest death rates. The deterministic-strategy death rate is comparable between included (17.8%) and excluded (14.9%) task types, confirming that prescreening does not select for death-prone tasks.

[2]Prescreening ensures that the counterfactual comparison is meaningful—task types where neither strategy makes progress provide no routing signal.

Task	n	Δ_{full}	$D\%_{\text{int}}$	$D\%_{\text{det}}$	D / I / N
id-life-stages-1	20	+47.50	55	10	9 / 0 / 11
lifespan-long	25	+44.00	56	16	11 / 1 / 13
lifespan-l-then-s	20	+40.80	45	10	8 / 1 / 11
inclined-plane	20	+36.00	55	20	7 / 0 / 13
freeze	20	+30.50	55	25	6 / 0 / 14
meas-melt-point	20	+30.05	55	25	6 / 0 / 14
id-life-stages-2	20	+30.00	50	20	6 / 0 / 14
find-animal	20	+26.65	50	25	5 / 0 / 15
grow-fruit	20	+25.10	45	20	6 / 1 / 13
chem-mix	20	+21.80	45	25	6 / 2 / 12
find-non-living†	40	+9.58	0	0	16 / 4 / 20
All (11 tasks)	**245**	**+29.60**	**42.9**	**16.3**	**86 / 9 / 150**

Table 2: Per-task decomposition of the death avoidance effect. D%: death rate for internal (int) and deterministic (det) strategies. D/I/N: episode labels (det-preferred / internal-preferred / no-preference). Sorted by Δ_{full} descending. † Only task with zero deaths under both strategies; n=40 because data was collected in two batches, with the second batch adding 20 episodes to increase statistical power for this zero-death outlier.

Task removed	r_{LOO}
id-life-stages-1	0.963
lifespan-long	0.969
lifespan-l-then-s	0.976
inclined-plane	0.969
freeze	0.971
meas-melt-point	0.972
id-life-stages-2	0.972
find-animal	0.970
grow-fruit	0.973
chem-mix	0.969
find-non-living	0.980

Table 3: Leave-one-out Pearson r for the death rate differential vs. Δ_{full} correlation. Full $r = 0.970$; LOO range $[0.963, 0.980]$. No single task type is a leverage point.

Rather, the key distinction is the *presence* of routing signal: all 11 included tasks have $\Delta_{\text{nd}} \neq 0$ (non-zero performance gap on surviving episodes), while 53% of excluded tasks show $\Delta_{\text{nd}} = 0$—neither strategy outperforms the other even when both survive, typically because both time out before making progress. `find-non-living-thing` serves as an internal counterexample: it has zero deaths under both strategies yet is included because it exhibits non-death routing signal ($\Delta_{\text{full}} = +9.58$ entirely from score differences), and deterministic is not always preferred (4 of 20 non-tie labels favor internal).

C Correlation Robustness

The Pearson $r = 0.9698$ reported in §3.1 is robust across multiple diagnostics. Spearman rank correlation is $\rho = 0.986$ ($p = 2.4 \times 10^{-8}$), confirming that the relationship is not driven by outliers or non-linearity. A nonparametric bootstrap (10,000 resamples) yields 95% CI $[0.952, 0.995]$.

D Router Training Details

We use Qwen3-1.7B for the router to avoid circular dependency with the Qwen2.5-7B data-generating model.

With gradient accumulation of 1 (effective batch size 4), routing performance is highly sensitive to the evaluation split, suggesting overfitting to specific train/validation partition boundaries. With gradient accumulation of 8 (effective batch size 32), step-level performance stabilizes across splits ($\sim$54% balanced accuracy). All reported results use gradient accumulation 8.

Under step-level evaluation with a fixed data split, cross-seed balanced accuracy ranges from 52.7% to 58.1%, but internal recall varies from 51.5% to 81.8%, reflecting threshold sensitivity. However, episode-level evaluation reveals that this apparent discrimination does not generalize: balanced accuracy drops to $47.1 \pm 2.2\%$ across 8 cross-split seeds (Appendix E). Deployed end-to-end, the trained router allocates 97.0% of episodes to deterministic computation with 17.6% death rate, and misrouting cost (death from internally-routed episodes) exceeds the benefit of correctly routing rare internal-preferred cases.

Evaluation methods. All results use *generation-based evaluation*: the model generates a routing label as text (`enable_thinking=False`), parsed for classification. We evaluate at two granularities: *step-level* (steps from the same episode may appear in both train and validation sets) and *episode-level* (entire episodes held out). Step-level accuracy ($54.1 \pm 5.4\%$) exceeds chance, indicating the router captures some signal, but episode-level accuracy ($47.1 \pm 2.2\%$, below chance) shows this does not generalize to unseen episodes. A data scaling ablation (Appendix E) rules out step-level data scarcity, pointing instead to a structural limitation: always-deterministic is trivially near-optimal given DAF = 90.4%. Both metrics are reported in the main text; the model-size ablation (Appendix I) uses step-level evaluation.

Hyperparameter	Classification Head	Implicit Router
Base model	Qwen3-1.7B	Qwen3-1.7B
Architecture	Linear + LoRA	SFT + LoRA
LoRA r / α	16 / 32	16 / 32
Learning rate	2e-5	2e-4
Optimizer	AdamW	AdamW
Epochs	3	3
Batch size (per device)	4	4
Gradient accumulation	8	8
Warmup ratio	0.1	0.1
Weight decay	0.01	0.01
Max sequence length	512	512
Seeds	{42}	{42, 123, 456, 789}

Table 4: Router training hyperparameters. Both architectures use the same base model with LoRA adaptation. The classification head degenerates across all configurations; the implicit router results are reported as 4-seed mean±std.

E Seed Robustness

We assess router generalization at two evaluation granularities: *step-level*, where steps from the same episode may appear in both train and validation sets, and *episode-level*, where entire episodes are held out. Both use cross-split evaluation (seeds $\{42, \ldots, 49\}$, 8 seeds total), where the data partition and training seed both vary. Table 5 reports per-seed results.

The contrast between step-level ($54.1 \pm 5.4\%$) and episode-level ($47.1 \pm 2.2\%$) evaluation reveals a generalization gap: the step-level signal likely reflects memorization of episode-specific context (enabled by same-episode leakage) rather than extraction of generalizable routing features. At episode level, the router performs below chance across all 8 seeds (range: 43.1%–49.6%). A data scaling ablation (Table 6) rules out step-level data scarcity as the cause: scaling from 25% to 800% of the original training set (32× range) yields no improvement, converging to the always-deterministic baseline (50%). The failure is structural: with DAF $= 90.4\%$ and all task types deterministic-majority (86:9 label ratio), always-deterministic trivially solves the episode-level routing problem, leaving no learnable signal for a step-level router.

Data scaling ablation. To test whether the episode-level generalization failure stems from data scarcity, we vary training set size from 25% to 800% of the original via paraphrase-based data augmentation (synonym substitution + template variation on step observations), holding the episode-level validation set fixed. Table 6 reports the results.

All scales hover at ∼50% episode-level balanced accuracy—equivalent to the always-deterministic baseline—with no monotonic improvement as data increases. This rules out step-level data scarcity as the explanation for the router's episode-level failure; episode-level minority-class diversity remains a potential confound that cannot be resolved without collecting more internal-preferred episodes. The ground-truth internal-preferred rate (9/245 $= 3.7\%$) means collecting 50 internal-preferred episodes would require approximately 1,350 paired episodes at current environment dynamics—a $5.5\times$ expansion with diminishing return on a 9.6% signal. The failure points instead to a structural limitation: when DAF $= 90.4\%$ and all task types are deterministic-majority (86:9 label ratio), the episode-level routing problem is trivially solved by always choosing deterministic, leaving no learnable signal for a step-level router to exploit.

Augmentation disproportionately increases the majority (deterministic) class: Det% rises from 89.6% at 100% to 94.2% at 200% and ∼97.7% at 800%. However, the complete collapse to majority-class prediction (internal recall $= 0.0$) occurs even at the smallest unaugmented scale (25%, ∼90%

Eval. Granularity	Seed	Bal. Acc. (%)
Step-level (same-episode leakage)	42	57.06
	43	63.99
	44	55.43
	45	56.14
	46	55.45
	47	50.08
	48	48.03
	49	59.30
	Mean±std	54.1 ± 5.4
Episode-level (held-out episodes)	42	48.1
	43	47.9
	44	49.6
	45	49.4
	46	48.5
	47	43.1
	48	45.4
	49	44.8
	Mean±std	47.1 ± 2.2
Always-det (= majority-vote)		50.0

Table 5: Per-seed router performance under cross-split evaluation (8 seeds, generation-based, `enable_thinking=False`). Step-level evaluation allows steps from the same episode in both train and validation sets (same-episode leakage). Episode-level evaluation holds out entire episodes; the resulting $47.1 \pm 2.2\%$ is below the 50% chance baseline. A data scaling ablation (Table 6) confirms this is structural rather than data-limited.

Scale	Train samples	Det%	Ep.-level Bal. Acc. (%)
25%	433	∼90	49.5 ± 0.3
50%	867	∼90	51.5
100%	1,735	89.6	49.9 ± 0.1
200%	3,470	94.2	50.0 ± 0.0
800%	13,880	∼97.7	51.0 ± 1.1
Always-det	—	—	50.0

Table 6: Data scaling ablation: episode-level balanced accuracy across training set sizes (25%–800% of original, 32× range). Internal recall = 0.0 at all scales (complete majority-class collapse). Det%: fraction of deterministic-preferred labels in training data. The 50% scale is a single seed; all others report mean±std across seeds.

det), confirming this is not an augmentation artifact. Balanced accuracy corrects for class imbalance by averaging per-class recall, so the metric itself is not biased by the skew.

Decomposition metrics are stable under bootstrap resampling (1,000 resamples of the 245-episode dataset): DAF = $90.6\% \pm 4.2\%$ (95% CI $[81.6\%, 98.8\%]$); Pearson $r = 0.975 \pm 0.011$ (95% CI $[0.952, 0.995]$). The death avoidance dominance finding is robust to both resampling and split perturbation.

Per-task-type evaluation breakdown. To test whether the router learns transferable routing strategies or merely task-type classification, we evaluate existing checkpoints (8 seeds) separately on each of the 11 task types. Of the 11 task types, 7 have *all-deterministic* labels (no internal-preferred episodes exist), making balanced accuracy undefined—the router has nothing to learn beyond the trivial always-deterministic default. For the remaining 4 task types where both classes are present, balanced accuracy ranges from 50.0% to 73.0%, but this apparent signal does not indicate cross-task-type routing transfer: the router is effectively classifying task types (which correlate with death risk) rather than extracting a generalizable routing strategy. This complements the data scaling ablation: routing signal does not transfer across task types because the signal *is* the task type—whether a given task carries death risk. The 4B capacity scaling result (Appendix I) suggests capacity is one factor, but even 4B's 59.0% episode-level accuracy does not demonstrate cross-task-type transfer.

P	DAF(P)	95% CI
0	7.5%	[−36.7, 83.1]
10	50.4%	[19.8, 94.5]
25	70.8%	[48.5, 97.1]
50	82.6%	[67.6, 98.4]
100*	90.4%	[81.5, 99.1]
200	95.0%	[90.0, 99.5]
500	97.9%	[95.8, 99.8]

Table 7: Sensitivity of DAF to death penalty magnitude P. DAF$(P) = 1 - \Delta_{\text{nd}}/\Delta_{\text{full}}(P)$, where $\Delta_{\text{full}}(P) \approx \delta \cdot P + \Delta_{\text{nd}}$, $\delta = 0.2653$ is the death-rate differential, and $\Delta_{\text{nd}} = 2.84$ (strict neither-died filtering). Bootstrap 95% CIs from 1,000 resamples. *Default setting.

F Penalty Sensitivity Analysis

The non-death score gap $\Delta_{\text{nd}} = 2.84$ is invariant across penalty levels—it depends only on surviving-episode performance, not on P. As P increases, Δ_{full} grows (16.33 at P=50, 29.60 at P=100, 56.13 at P=200) while Δ_{nd} remains fixed, mechanically increasing DAF.

Mechanism boundary: death specificity. The $r = 0.97$ correlation between per-task death-rate differential and Δ_{full} is specific to catastrophic death (score = −100). Broader failure definitions dissolve the correlation entirely: defining "failure" as any non-positive score (T=0) yields $r = -0.13$ ($p = 0.70$), because non-death episodes show near-zero performance differential between strategies ($\Delta_{\text{nd}} = 2.84$). Table 8 quantifies this specificity across penalty magnitudes.

P	r	p	Sig. (α=0.05)
10	−0.08	0.813	No
25	0.51	0.108	No
50	0.86	0.0006	Yes
100*	0.97	8×10^{-7}	Yes
200	0.99	1×10^{-9}	Yes

Table 8: Pearson r between per-task death-rate differential and $\Delta_{\text{full}}(P)$ across penalty magnitudes. The correlation is non-significant at low penalties and emerges at $P \geq 50$, reflecting the transition from non-death-dominated to death-dominated score gaps. *Default ScienceWorld setting.

This sharpens our central finding: death is not merely the dominant component of routing signal—it is effectively the *only* component that generates differential routing signal in ScienceWorld. At low penalties, the death contribution is small and Δ_{full} reduces to $\Delta_{\text{nd}} \approx 2.84$, which is near-uniform across task types (mean non-death gap = −1.19

excluding the `find-non-living-thing` outlier); consequently, death-rate differential has no predictive power. As P increases, the death term dominates Δ_{full}, and the correlation sharpens because death rates *are* the performance gap. The DAF values in Table 7 increase monotonically with P, but this is a mathematical consequence of the decomposition (DAF $= \delta P/(\delta P + \Delta_{\text{nd}})$), not evidence of mechanism robustness.

G Cross-Model Verification

To test whether the death-avoidance finding generalizes beyond Qwen2.5-7B-Instruct, we replicate the counterfactual analysis with Llama-3.1-8B-Instruct on 7 task types (N=70 episode pairs, 10 episodes per task). The results are strongly consistent with the Qwen findings: 24 episodes (34.3%) are deterministic-preferred, **0** (0%) are internal-preferred, and 46 (65.7%) show no preference. The internal strategy dies in 58.6% of episodes versus 24.3% for deterministic (death-rate differential $+34.3\%$), compared to $+26.6\%$ for Qwen2.5-7B. Direction consistency is perfect: all 7 task types show higher internal than deterministic death rates (7/7, compared to 10/11 for Qwen). The larger death-rate differential ($+34.3\%$ vs. $+26.6\%$) and the absence of any internal-preferred episodes (0 vs. 9 for Qwen) suggest that Llama-3.1-8B is more vulnerable to hallucination-induced death, amplifying the death-avoidance effect. These results confirm that the death-avoidance dominance is not specific to the Qwen model family.

H Failure Mode Analysis

The most common failure mode for the internal strategy is state hallucination: the model predicts that an action succeeded (e.g., "the water is now boiling") when the environment state did not change, leading to cascading errors that trigger agent death. The deterministic strategy avoids this by receiving ground-truth observations, but still dies in 16.3% of episodes—primarily from selecting inherently dangerous actions (e.g., eating toxic substances) rather than from state misestimation.

I Ablation Studies

Capacity scaling. We scale the implicit router from Qwen3-0.6B to Qwen3-4B and evaluate at both step and episode level (Table 9).

Model	Step Bal. Acc. (%)	Ep. Bal. Acc. (%)
Qwen3-0.6B	52.8	50.67 ± 2.62
Qwen3-1.7B†	54.1	47.10 ± 2.20
Qwen3-4B	53.5	$\mathbf{58.96 \pm 1.94}$
Always-det	50.0	50.0

Table 9: Capacity scaling: step-level and episode-level balanced accuracy. Step-level accuracy is flat across scales (52.8–54.1%). At episode level, 4B first breaks the random baseline (58.96%, 95% CI $[55.1\%, 62.8\%]$). † 1.7B episode-level figure comes from the main experiment (§3.3) with a different training configuration (8 cross-split seeds vs. 4 seeds for 0.6B/4B); direct comparison across scales should be interpreted with this caveat.

Step-level balanced accuracy is essentially flat across scales (52.8–54.1%), indicating that step-level discrimination is not capacity-sensitive. At episode level, however, the 4B model first breaks the random baseline ($58.96 \pm 1.94\%$, 95% CI $[55.1\%, 62.8\%]$), suggesting that model capacity is one limiting factor for episode-level generalization. Nevertheless, even 4B achieves only $\sim$59%—far from a practically useful router—and the dominant explanation remains that death avoidance constitutes a task-type-level binary signal that a step-level router cannot exploit efficiently.

Reasoning depth. An ablation comparing chain-of-thought reasoning against direct-answer generation on 6 task types (n=120 episodes per condition, 6 tasks $\times$ 20 episodes per task) found no significant difference (Mann–Whitney $p > 0.26$, $|d| \leq 0.31$) and near-identical death rates (41.8% vs. 42.5%), confirming that reasoning depth does not meaningfully alter internal-mode outcomes.

J Cross-Environment Analysis: ALFWorld

We evaluate on ALFWorld (Shridhar et al., 2021), a text-based household environment with 6 task types (json_2.1.1 split, valid_seen, 135 games, max 50 steps, binary success metric), using Qwen2.5-7B-Instruct. Unlike ScienceWorld, ALFWorld has no death penalty—failure means exhausting the step budget. The two modalities differ in *information access*: the internal strategy receives only free-text observations, while the deterministic strategy additionally receives admissible actions at each step. This is an information asymmetry rather than a computation mode switch.

The success rate gap is large: 27.4% (deterministic) vs. 3.0% (internal), reflecting the 7B model's near-inability to solve household tasks without

admissible-action guidance. The label distribution (Table 10) shows 35 deterministic-preferred (25.9%), 2 internal-preferred (1.5%), and 98 no-preference (72.6%), of which 96 are both-fail. The deterministic advantage stems entirely from the information gap rather than death avoidance.

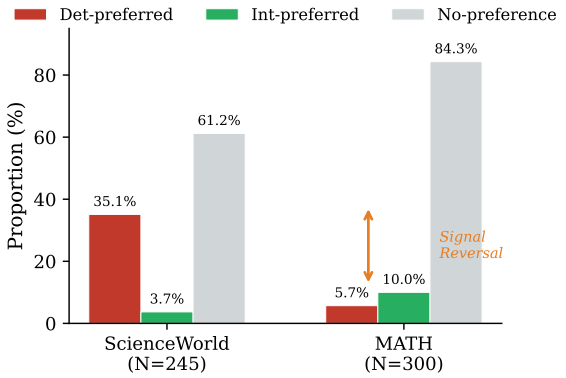


Figure 3: Label distribution reversal across environments. In ScienceWorld (with -100 death penalty), deterministic-preferred labels dominate (86:9). In MATH (no death penalty), the pattern inverts (17:30), with 84.3% no preference.

Environment	N	Det%	Int%	NoPref%	Death?
ScienceWorld	245	35.1	3.7	61.2	Yes (-100)
MATH	300	5.7	10.0	84.3	No
ALFWorld	135	25.9	1.5	72.6	No†

Table 10: Cross-environment label distribution. † ALFWorld's deterministic advantage stems from action-space pruning, not death avoidance.